TerraMind AI — Final Research Report

TerraMind AI: Multimodal Fusion and Agricultural Decision Support

Final experimental status: the three fusion architectures are implemented, but no valid paired text dataset or trained fusion checkpoints are available. Therefore, this report does not claim a three-way fusion result. The measured classifier comparison is between the image-only EfficientNet-B0 and one separate, one-epoch class-weighted candidate.

Abstract

TerraMind AI is a decision-support prototype that combines a PlantWild image classifier with retrieval-augmented agricultural guidance. This report consolidates persisted measurements for the production EfficientNet-B0, a class-weighted experimental candidate, and a ten-query source-retrieval evaluation. The matched duplicate-free official test comparison measured 60.68% accuracy and 57.40% macro-F1 for the baseline, versus 60.93% and 57.90% for the candidate. The candidate remains experimental. Early, intermediate, and late fusion implementations are untrained because PlantWild class labels and class-level prompts are not independent per-image text and would leak the target. RAG retrieval returned 251 points sourced from 218 documents; on ten manually labeled questions, precision@5 was 0.18. Local integration tests passed, but hosted services were not verified. Outputs are guidance and model predictions, not confirmed diagnoses.

1. Problem statement and objectives

Plant disease recognition in uncontrolled imagery is difficult because of visual variation within a disease class and similarity between classes. The project objective is to provide image-based candidate predictions and cite relevant agricultural knowledge while preserving uncertainty. The research objective is to compare an image-only baseline with three fusion mechanisms without test leakage or fabricated paired observations. The fusion comparison remains blocked by the data requirement described below.

2. System architecture

The Next.js interface calls a FastAPI service. The backend lazily loads the production EfficientNet-B0 for image inference, exposes explicit model availability, and uses a source-backed retrieval pipeline with TF-IDF embeddings and a local persistent fallback (Qdrant is optional). A supervisor routes selected requests to existing specialist workflows and records which agents ran. Optional LLM generation is server-configured; when no provider is configured, the backend returns an evidence-template response with citations. Supabase handles user identity and durable records when reachable; local report/review files are a fallback, not cross-device persistence. The frontend displays model scores separately from test-set metrics and exposes uncertainty/review paths.

3. Dataset and split protocol

The repository’s PlantWild v1 material describes 18,542 images across 89 classes and supplies official split metadata. The project artifacts record 13,045 training, 1,820 validation, and 3,677 test rows. Content-duplicate filtering excluded 69 validation and 132 test entries and yielded 1,751 validation and 3,545 test eligible images for the duplicate-free comparison. Model selection uses validation macro-F1; the comparison artifact records that test data was not used for selection.

4. Image-only EfficientNet-B0 baseline

The unchanged production checkpoint is

`models/archive/2026-09-09_efficientnet_b0_official_v3/best_model.pth`, with 89 outputs and 224-pixel input. Its companion training artifact records seed 42, 20 epochs, and best validation macro-F1 56.5659% at epoch 20. This is the serving checkpoint; its stored result reports accuracy 60.0218% and macro-F1 56.6969%.

Reconciliation of production metrics

The stored result artifact does not record its test sample count or enough evaluation-protocol detail to reproduce the score. Independent evaluation of the named checkpoint on all 3,677 official test rows measured 59.72% accuracy and 56.50% macro-F1. A separate duplicate-free evaluation on 3,545 rows measured 60.68% and 57.40%. These values differ because the stored score and independent evaluation are not demonstrably the same evaluation record/protocol. We preserve all three and use only the duplicate-free baseline for the matched candidate comparison; we do not overwrite or average the records.

5. Early, intermediate, and late fusion methods

The repository implements reusable classifiers and training/evaluation interfaces:

- **Early fusion:** concatenate projected image and text features before the classification head.
- **Intermediate fusion:** combine projected encodings through a learned gate.
- **Late fusion:** combine image and text class probabilities, with modality weights selected using validation data.

All three are **implemented but untrained and unavailable for inference**. They have no test accuracy, F1, confusion matrix, parameter/inference measurements, or checkpoint to report. Their blank metrics in the generated CSV/JSON mean unavailable, not zero.

6. Text modality and leakage controls

PlantWild provides class names and GPT-generated descriptions per class, not independent farmer observations or captions paired to each image. The existing multimodal preparation that uses class labels is class-conditioned; using it as model text would reveal the target. The fusion pipeline requires provenance-bearing per-image text, rejects label/target columns and class-name content, checks paths and split alignment, and fits the vocabulary on training text only. These checks cannot authenticate a curator’s provenance assertion. Valid paired data must be collected or supplied and also be available at inference before fusion training can proceed.

7. Experimental setup and image-only candidate

The candidate is an EfficientNet-B0 initialized compatibly with the production model and trained for one epoch using class-weighted loss on the official training split. It was selected on validation macro-F1 (58.2429% versus baseline 56.8672% on the duplicate-free validation subset). The comparison artifact records a 642.8-second filesystem interval from config creation to results write; this is not a trainer-only timer. It was evaluated on the same 3,545 duplicate-free official test examples as the baseline. Test metrics were not used to select it. The experimental checkpoint remains separate and is not promoted.

8. Measured model comparison

| Model/checkpoint | Evaluation subset | N | Accuracy | Macro-P/R/F1 | Weighted-F1 | Top-3 | Parameters |

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Size (MB) | CPU ms/image |
|---|---|---:|---:|---:|---:|---:|---:|---:|---:|
| Production EfficientNet-B0 v3 | Duplicate-free official test | 3,545 | 60.68% | 57.80%/59.28%/57.40% | 60.32% | 80.45% | 4,121,557 | 16.78 | 35.45 |
| One-epoch class-weighted candidate (not promoted) | Same duplicate-free official test | 3,545 | 60.93% | 58.00%/59.41%/57.90% | 60.94% | 80.82% | 4,121,557 | 16.78 | 32.34 |
| Early fusion | Not evaluated | — | Unavailable | Unavailable | Unavailable | Unavailable | Unavailable | Unavailable |
| Intermediate fusion | Not evaluated | — | Unavailable | Unavailable | Unavailable | Unavailable | Unavailable | Unavailable |
| Late fusion | Not evaluated | — | Unavailable | Unavailable | Unavailable | Unavailable | Unavailable | Unavailable |

On this finite duplicate-free subset, the candidate’s measured delta was +0.254 percentage points accuracy and +0.505 points macro-F1. This is a single candidate run, not evidence of robust improvement or external-field performance, and promotion was not automatic. The candidate’s weighted-F1 was 60.94%, compared with baseline 60.32%; top-3 was 80.82% versus 80.45%. The comparison CSV/JSON identifies the checkpoint paths, subset, n, and metric sources.

9. Per-class error analysis

The baseline error analysis is calculated from actual duplicate-free test predictions. The ten lowest-F1 classes are:

Class	Support	Precision	Recall	F1
bell pepper leaf spot	23	0.083	0.087	0.085
tomato bacterial leaf spot	54	0.643	0.167	0.265
bean leaf	47	0.344	0.234	0.278
tomato septoria leaf spot	40	0.323	0.250	0.282
apple mosaic virus	34	0.381	0.235	0.291
cherry powdery mildew	18	0.333	0.278	0.303
ginger sheath blight	14	0.294	0.357	0.323
tomato mosaic virus	36	0.474	0.250	0.327
cabbage alternaria leaf spot	23	0.270	0.435	0.333
peach leaf	31	0.314	0.355	0.333

Most frequent measured confusions include:

Actual	Predicted	Count
grape downy mildew	grape black rot	15
apple leaf	blueberry leaf	11
corn gray leaf spot	corn northern leaf blight	11
grape downy mildew	grape leaf spot	10
squash leaf	cucumber leaf	9
squash powdery mildew	cucumber powdery mildew	9
potato early blight	potato late blight	8

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tomato early blight	tomato septoria leaf spot	8
tomato mosaic virus	tomato yellow leaf curl virus	8
apple leaf	cherry leaf	7

The split’s class-count artifacts are included under the baseline and candidate error-analysis directories. The evaluation script identified no dedicated healthy-class IDs in the current mapping, so a separate healthy-versus-diseased score is unavailable. Per-class CSVs and 89×89 confusion matrices are included alongside this report. Temperature scaling was fitted on 1,751 duplicate-free validation examples; the baseline calibrated duplicate-free test ECE was 1.81%, compared with raw ECE 10.58%. Calibration changes confidence, not predicted model accuracy.

10. RAG architecture and retrieval evaluation

The pipeline preprocesses the question, retrieves and ranks chunks, then returns source IDs/titles and citations with either an optional configured provider or a transparent evidence template. Diagnosis class context is retrieval context only and low-confidence output remains explicitly unconfirmed. The current measured index reports **218 source documents and 251 vector points** using `local-memory`; category point counts are `{"diseases": 91, "treatment": 58, "fertilizer": 34, "plant care": 34, "agriculture": 34}`. A source can produce more than one chunk.

On the existing manually labeled ten-query evaluation (two queries per category), measured hit rate/recall@5 was **0.90**, precision@5 **0.18**, MRR **0.783**, and citation-presence **0.90**. Precision is weak: only 18% of the top-five retrieved passages were judged relevant under this small label set. This is a smoke evaluation, not a broad benchmark. Full cases and labels are retained in `rag_retrieval_evaluation.json`/`.csv`.

11. Supervisor and specialist agents

The supervisor selects existing specialists for request context. The diagnosis workflow records Vision, Diagnosis, Research/RAG, Treatment, Fertilizer, Care, and Report execution events when applicable; chat can route an agricultural question to a relevant specialist. API tests inspected actual returned event records, and unit tests covered category routing, citations, no-evidence responses, provider failure fallback, and local vector-store persistence. Agent guidance is not a model diagnosis, and citations are retained from retrieval evidence.

12. Application integration

FastAPI serves single/batch image inference, model registry/status, analytics artifacts, RAG/chat, agent workflows, and ownership-checked history/review routes. The Next.js production build completed locally and reads backend responses through configured rewrites. The candidate is explicitly unpromoted; all three fusion choices remain unavailable. The service uses confidence calibration when a checkpoint-matched validation sidecar is available and preserves raw scores for display.

13. Verification and deployment

Verification record: `local_checks_passed_hosted_services_unverified`. Local tests/builds are not hosted-service checks. The local Windows API process measured approximately Unavailable MiB RSS at startup after the configured single-thread limit. Render Free advertises 512 MB memory and an ephemeral filesystem; this local process leaves little headroom, and no Render/Linux fit or live endpoint was verified. Supabase connectivity and a live LLM provider were not tested. The local API reported LLM

unconfigured and uses evidence-template generation.

14. Limitations and ethical considerations

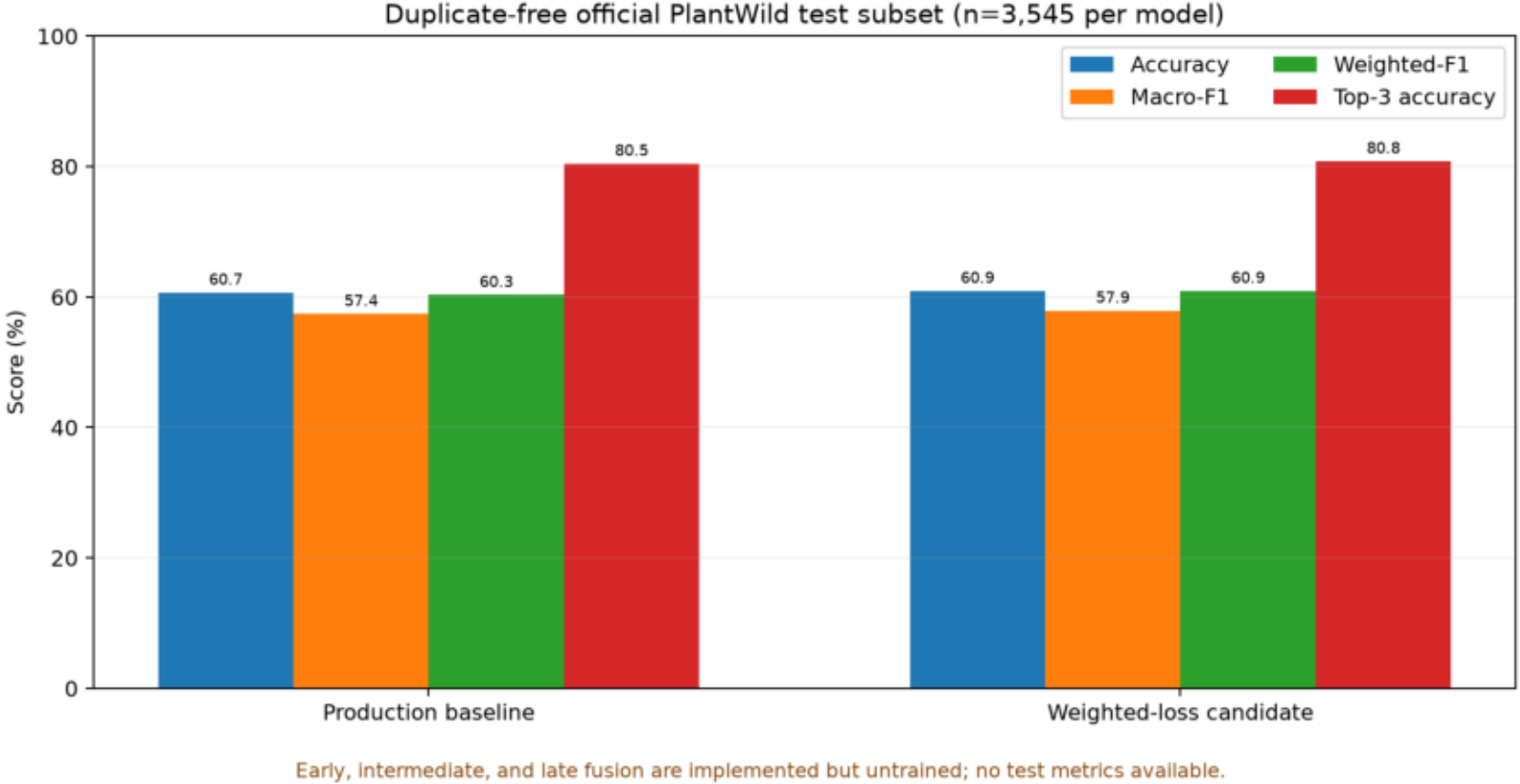
PlantWild evaluation does not establish performance on a particular farm, cultivar, camera, or region. The class prediction is a candidate label; it is not a confirmed agricultural diagnosis. Users should inspect source-backed guidance and seek qualified local expertise, particularly for uncertain images or pesticide decisions. The ten-query retrieval set is too small to establish category-wide retrieval quality. Fusion comparisons await genuinely paired text. The one-epoch candidate has limited training and must not replace the production checkpoint without a documented, protocol-matched evaluation and review.

15. Conclusion

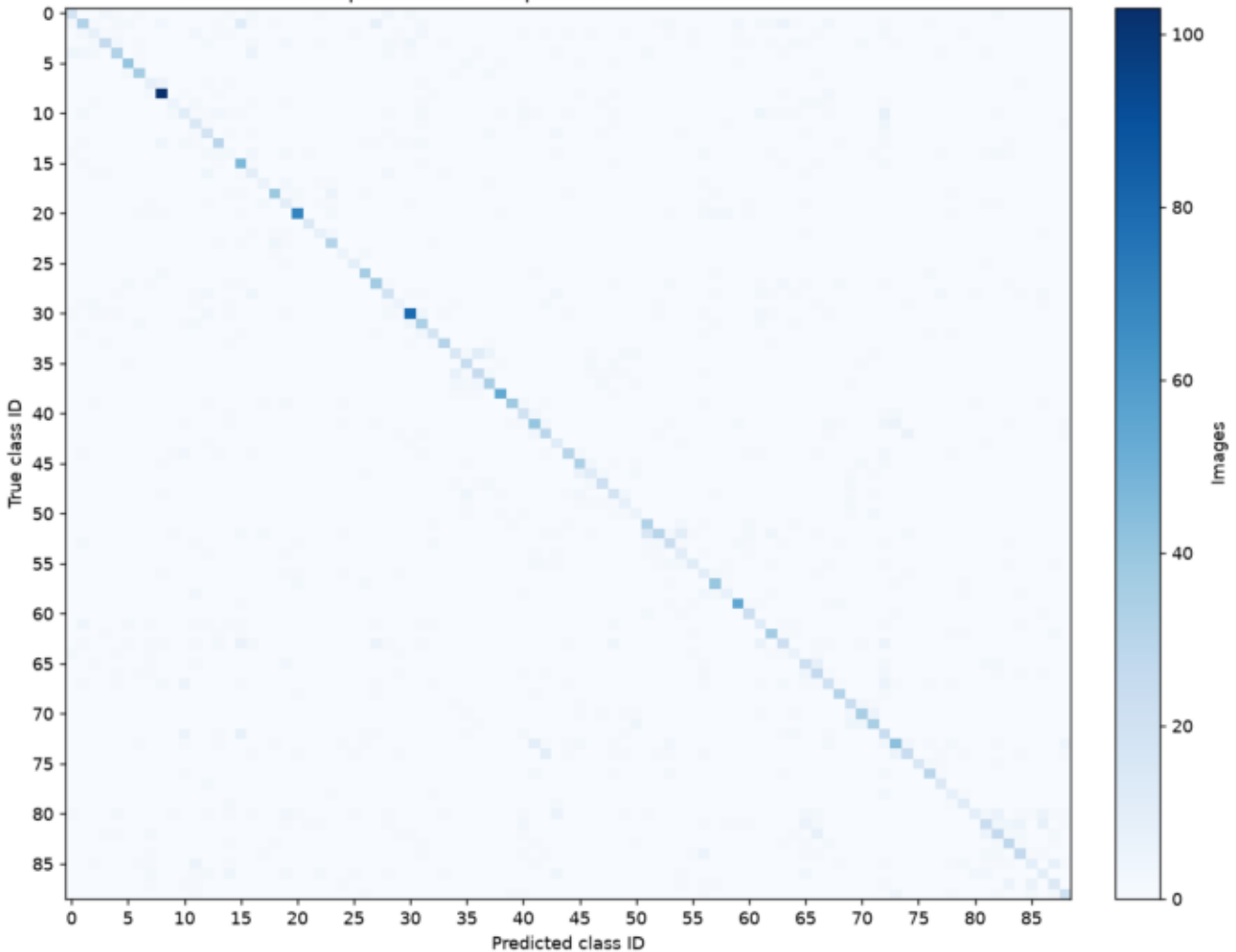
This release delivers an operational image-only decision-support path and a measured, cited RAG prototype. The best matched test result in the current saved comparison is the separate candidate at 60.93% accuracy and 57.90% macro-F1, but it remains experimental. The three fusion models are architectures only: their results are unavailable until non-label-derived text paired to each image is supplied. The stored production result discrepancy is retained and documented rather than hidden. External deployment, persistent hosted storage, and provider behavior remain unverified.

References and repository evidence

1. Wei, T., Chen, Z., Huang, Z., & Yu, X. (2024). *Benchmarking In-the-wild Multimodal Disease Recognition and A Versatile Baseline*. arXiv:2408.03120. https://arxiv.org/abs/2408.03120
2. Tan, M., & Le, Q. (2019). *EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks*. ICML, PMLR 97, 6105–6114. https://proceedings.mlr.press/v97/tan19a.html
3. Singh, D. et al. (2020). *PlantDoc: A Dataset for Visual Plant Disease Detection*. ACM India Joint International Conference on Data Science and Management of Data. https://arxiv.org/abs/1911.10317. PlantDoc is contextual literature, not paired text used in these experiments.
4. Dataset documentation and official split files: `data/plantwild/plantwild/readme.txt`, `train_split.txt`, `val_split.txt`, `test_split.txt`, `classes.txt`.
5. Source-backed RAG examples (the original JSON records, with provenance retained): `data/knowledge_base/diseases/apple/apple_black_rot.diseases.json`; `data/knowledge_base/treatment/tomato/tomato_late_blight.treatment.json`; `data/knowledge_base/fertilizer/tomato/tomato.fertilizer.json`; `data/knowledge_base/plant_care/cucumber/cucumber.plant_care.json`; `data/knowledge_base/agriculture/corn/corn.agriculture.json`.
6. Measured artifacts: `outputs/experiments/efficientnet_b0_comparison/comparison.json`; `outputs/experiments/efficientnet_b0_error_analysis/metrics.json`; `outputs/experiments/efficientnet_b0_weighted_loss_v1_error_analysis/metrics.json`; `outputs/research/rag_retrieval_evaluation.json`.


Duplicate-free official PlantWild test subset (n=3,545 per model)
Accuracy
Macro-F1
Weighted-F1
Top-3 accuracy
Score (%)
100
80
60
40
20
0
60.7
57.4
60.3
80.5
60.9
57.9
60.9
80.8
Production baseline
Weighted-loss candidate
Early, intermediate, and late fusion are implemented but untrained; no test metrics available.

confusion matrix baseline duplicate free



confusion matrix candidate duplicate free

